Would you ship on this simulation? A backtest for A/B decisions

TL;DR: Customers use simulations to decide what to ship, but the usual accuracy check (do simulated answers look like real ones?) can miss whether the simulation gets the *decision* right. I built a backtest on 17 real experiments that scores any simulator on the ship decision. Plugging in Simile's model would show where its confidence labels can be trusted for ship calls.

The customer question

A PM has two versions and asks a simulation: *should I ship B?* Before acting, they need to know whether that answer is safe to act on, or whether they still need real people.

What I built

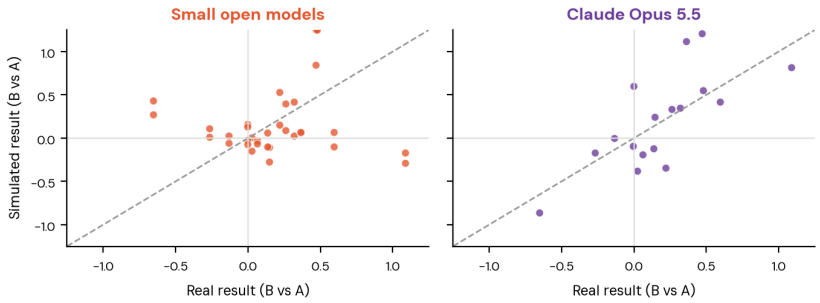
A backtest made of 17 published experiments with known real answers. For each one it checks whether a simulator picks the same winner, makes the same ship call, and how many real people its answers are worth. It works with any simulator: swap one in and rerun.

What it shows on off-the-shelf models

- **Looking realistic isn't following reality.** Small models gave realistic-sounding answers, but whether B beat A in the simulation had little to do with real life.
- **Pointing the right way isn't a reliable ship call.** Claude Opus 5.5 picked the right winner in 10 of 13 experiments that had one, yet no simulator made the ship / don't-ship call more reliably than a team that always ships B.
- **A few real people on top didn't help.** Mixing simulated answers with a small real sample didn't reduce how many real people were needed.

Lesson: realistic answers and the right direction aren't the same as decision-ready, so decisions need their own check.

Looks realistic vs follows reality



Next step for Simile

Run Simile's simulator through the same backtest. Check whether results labeled high confidence are the ones that get the ship call right. Then express confidence in decision terms: **"worth about N real people"** and **"you still need about M"**. The backtest grows with every experiment added, for example Simile's own validation studies.

Tommy Yu | github.com/tommuthecow/effect-fidelity

Details: small models are Qwen2.5-1.5B and Llama-3.2-1B; Claude: Claude Opus 5.5, one quick run (25 simulated people per version). Correlation of simulated vs real differences: small models 0.05 to 0.19, Claude Opus 5.5 0.73. Ship call (ship B / keep A / no difference, from a 95% interval) wrong 47% to 64% of the time for every simulator vs 41% for always shipping B and 54% for 25 real people. Mixing used prediction-powered inference (about 1.0x worth; Claude tested with 20 real people per version). Chart: dashed line = simulation matches reality. Prior work: Huang et al. 2025, arXiv 2502.17773; Broska et al. 2025, The Mixed Subjects Design, Sociological Methods & Research. Methods & audit in repo.